

# The Human Baseline for Strong-Lens Grading: Inter-Grader Reliability and Grade-Stratified Confirmation Rates for the DESI Legacy Surveys Lens Search

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*What “as good as a human grader” must mean, measured from published data alone: the program’s first inter-grader agreement statistics, its first confirmation-purity-per-grade table, and the concrete parity targets they set for any machine grader*

## Abstract

Any claim that an automated system grades strong-lens candidates “at human level” is empty until human level is a number. For the DESI Legacy Surveys search (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024) that number has never been published: final A/B/C grades are a two-senior-grader consensus, and no paper in the series (or, to our knowledge, in any survey’s lens search) reports a chance-corrected inter-rater statistic for expert lens grading. We recover the missing baseline from published data alone, in two independent ways. *Reliability*: the Huang et al. (2021) VizieR catalog preserves, for each of 1,312 accepted candidates, the two graders’ score average and absolute difference — enough to reconstruct every unordered pair of integer scores. The two graders gave the same 1–4 score on only 55.4% of accepted candidates (95% CI [52.7, 58.0]), for a symmetrized quadratic-weighted kappa of 0.42 [0.37, 0.47]; on grade-A candidates — the search’s highest-confidence class — both graders said “4” only 46.5% of the time, and on 3.7% of *accepted* candidates one grader had voted at the reject level. Because the catalog is truncated at acceptance, each of these is an *upper* bound on full-pool agreement. Against independent expert teams grading the same candidates the ceiling is lower still ( $\kappa_{\text{qw}} = 0.29$  vs. the SuGOHI committee,  $n = 103$ ; 0.17 vs. the Euclid Q1 panel,  $n = 24$ ). *Validity*: assembling every published per-target follow-up outcome for the program’s candidates — HST SNAP (Huang et al., 2026), DESI EDR spectroscopy (Huang et al., 2025), Keck/NIRES (Agarwal et al., 2025), VLT/MUSE (Lin et al., 2025), AGEL (Barone et al., 2025; Tran et al., 2022), HSC×DESI (Shu et al., 2025), and SuGOHI spectroscopy (Sonnenfeld et al., 2018) — yields the program’s first grade-stratified confirmation table: of 162 followed-up graded candidates, 130 are decided, with purity  $79/83 = 95.2\%$  [88.9, 98.4] for grade A,  $23/25 = 92.0\%$  for B, and  $20/22 = 90.9\%$  for C — statistically indistinguishable across grades on this (curated, best-first) follow-up mix, and pointedly *not* 100% for grade A: four grade-A candidates are spectroscopically refuted, all arcs at the wrong or same redshift. Together these results replace “trained humans reliably assign A/B grades” with measured quantities, and they define the two pre-registrable targets any machine grader must meet: an agreement target (match one grader’s  $\kappa_{\text{qw}}$  against the consensus, bounded by [0.29, 0.78]) and a truth target (non-inferior confirmed-vs-refuted discrimination on the same follow-up set). The machinery and caveats for both are laid out here.

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# 1 Why a human baseline

The DESI Legacy Surveys strong-lens search grades neural-network recommendations by expert visual inspection: candidates surviving a probability cut are scored by team members and published with grades A (“high level of confidence”), B, or C. Downstream uses — follow-up target selection, purity claims, training labels for the next finder, and any assessment of an automated grader such as LENSJUDGE — treat the grade as a stable quantity. Part I of the LENSJUDGE report measured how well an automated grader reproduces the consensus grade at DESI resolution and found  $\kappa \approx 0$ ; but it also found that *independent expert teams* agree with the DESI consensus only at  $\kappa_{\text{qw}} \approx 0.3$ , and the literature reports high intra-expert inconsistency in every controlled experiment (Cañameras et al., 2021; Euclid Collaboration et al., 2025; Petrillo et al., 2019; Rojas et al., 2023). What has been missing is the number for *this* program’s own graders: no inter-grader agreement statistic and no confirmation rate conditioned on grade has ever been published, here or (to our knowledge) for any survey’s expert lens grading. This report supplies both, from published data alone, with no new grading campaign.

Two distinct questions hide inside “as good as a human”: *reliability* (do graders agree with each other?) and *validity* (do the grades predict what spectroscopy and high-resolution imaging later decide?). We measure both, because they set different machine targets: a machine can only be asked to agree with the consensus as well as a human does (§3–4), and it can be asked to discriminate real from false candidates at least as well as the grades do (§5).

## 2 The recovered two-grader data

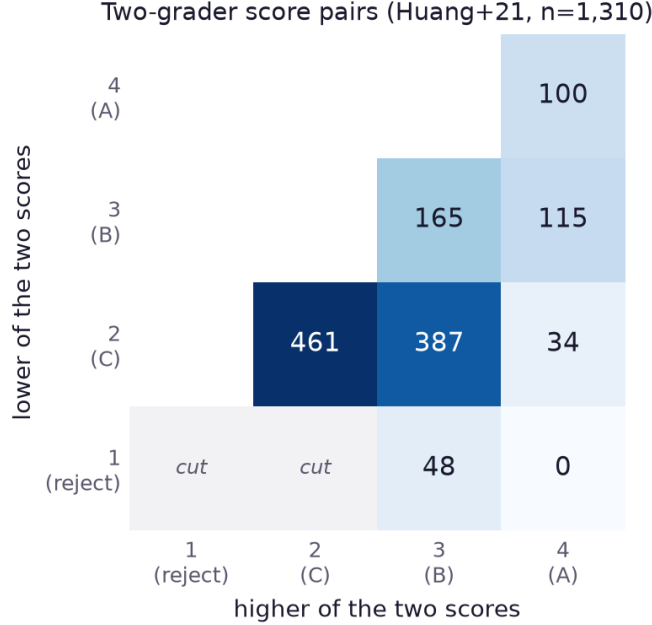
In Huang et al. (2021) (“Paper II”), two graders each assigned an integer score 1–4 to every candidate surviving triage; the published catalog records the pair’s *average* (`Score`  $\in [2.0, 4.0]$ , since only averages  $\geq 2.0$  were accepted) and, crucially, the *absolute difference* (`delSc`  $\in \{0, 1, 2\}$ ) — columns preserved in the Vizier table J/ApJ/909/27/cand but absent from every other release of the series. The unordered pair of integer scores is exactly  $\{\text{Score} - \text{delSc}/2, \text{Score} + \text{delSc}/2\}$ ; grader *identity* is lost, so all statistics below treat the raters as interchangeable. Parity of  $2 \times \text{Score}$  and `delSc` validates the reconstruction row by row: 1,310 of 1,312 rows are consistent with two integer scores (the two exceptions, DESI-094.5639+50.3059 and DESI-227.9362+06.5090, carry impossible combinations — data-entry glitches — and are excluded).

Letter grades follow from the average (A:  $\geq 3.5$ ; B: 3.0; C: 2.0–2.5), which checks out exactly against the published counts (216 A = 100 pairs of  $\{4, 4\}$  +115 of  $\{3, 4\}$  +1 glitch; 199 B; 897 C). Figure 1 shows the full joint distribution.

## 3 Inter-grader reliability

Table 1 gives the program’s first inter-grader agreement statistics (2,000-sample bootstrap over candidates, seed 2026). Three facts carry the story:

- **Exact agreement is a coin flip plus a little.** The two graders gave the same integer score on 55.4% of accepted candidates; 38.3% differ by one step and 6.3% by two. The symmetrized quadratic-weighted kappa is 0.42 (“moderate”), identical to Krippendorff’s interval  $\alpha$  as expected for two coders. For comparison, Rojas et al. (2023) measured 73% *intra*-rater repeat consistency — an individual agrees with the same image shown twice more often than these two experts agree with each other.



**Figure 1:** The two graders’ unordered score pairs for all 1,310 reconstructable accepted candidates in [Huang et al. \(2021\)](#). Cells marked *cut* cannot appear in the published catalog (pairs averaging below 2.0 were not accepted), which is why every statistic in §3 is an upper bound on full-pool agreement. Note the 48 candidates scored  $\{1, 3\}$ : one grader at reject level, the other at grade-B confidence, on the same cutout.

- **The top grade is the least internally stable.** Within published grade A, both graders voted “4” only 46.5% [40.2, 53.2] of the time; the majority of A’s are a  $\{3, 4\}$  split. (Grade B’s higher within-grade agreement, 82.9%, is partly mechanical: a  $\{3, 4\}$  pair averages to A and a  $\{2, 3\}$  pair to C, so B can only arise as  $\{3, 3\}$  or the rare  $\{2, 4\}$ .)
- **Acceptance can hinge on one vote.** 3.7% of *accepted* candidates (48 of 1,310) carry a  $\{1, 3\}$  pair — one grader would have rejected outright. And the truncation cut means every pair in which *both* graders were low is invisible: these numbers are ceilings.

The last row block of Table 1 contains the number a machine should be scored against for an *agreement* claim: an individual trained grader matches the published consensus at  $\kappa_{\text{qw}} = 0.776$  [0.757, 0.793] — and even this is inflated, since each grader contributes half of that consensus. A machine that reaches  $\kappa_{\text{qw}} \approx 0.78$  against the published grades is *indistinguishable from a member of the grading team*; one that reaches the two graders’ mutual 0.42 already agrees with a Huang-group grader as well as the two graders agree with each other.

## 4 Independent teams agree far less

The Paper II pair shares a rubric, a rendering, and a screen. Independent expert teams grading the *same* candidates under their own protocols set a lower — and for machine-parity purposes, arguably fairer — ceiling. Re-deriving the LENSJUDGE Part I crossmatch from its now fully staged inputs reproduces: DESI consensus vs. the SuGOHI  $\sim 9$ -grader committee ([Sonnenfeld et al., 2018](#)),  $\kappa_{\text{qw}} = 0.288$  [0.121, 0.444],  $n = 103$ , exact agreement 38.8%; DESI vs. the Euclid Q1  $\sim 10$ -expert panel ([Euclid Collaboration et al., 2025](#)),  $\kappa_{\text{qw}} = 0.165$ ,  $n = 24$ . The literature concurs: only 30%

| statistic (Paper II, $n = 1,310$ accepted candidates)                                                              | value | 95% CI         |
|--------------------------------------------------------------------------------------------------------------------|-------|----------------|
| exact score agreement $P(\text{delSc} = 0)$                                                                        | 0.554 | [0.527, 0.580] |
| one-step disagreement                                                                                              | 0.383 | [0.357, 0.411] |
| two-step disagreement                                                                                              | 0.063 | [0.050, 0.076] |
| one grader at reject level (among accepted)                                                                        | 0.037 | [0.026, 0.047] |
| symmetrized $\kappa_{\text{qw}}$ (scores 1–4)                                                                      | 0.420 | [0.371, 0.468] |
| Krippendorff $\alpha$ (interval)                                                                                   | 0.420 | [0.371, 0.468] |
| individual grader vs. published consensus $\kappa_{\text{qw}}$<br>(upper bound: each grader is half the consensus) | 0.776 | [0.757, 0.793] |
| both graders “4” within grade A                                                                                    | 0.465 | [0.402, 0.532] |
| exact agreement within grade B                                                                                     | 0.829 | [0.776, 0.881] |
| exact agreement within grade C                                                                                     | 0.515 | [0.480, 0.547] |

**Table 1:** Inter-grader reliability reconstructed from the [Huang et al. \(2021\)](#) Vizier Score/delSc columns. All quantities are upper bounds on full-pool agreement (truncation at acceptance). Code: `parity/intergrader_stats.py`.

| grade | followed | confirmed | refuted | lens-z only | pend./inc. | decided | purity (Jeffreys 95%) |
|-------|----------|-----------|---------|-------------|------------|---------|-----------------------|
| A     | 99       | 79        | 4       | 10          | 6          | 83      | 0.952 [0.889, 0.984]  |
| B     | 30       | 23        | 2       | 4           | 1          | 25      | 0.920 [0.767, 0.983]  |
| C     | 33       | 20        | 2       | 7           | 4          | 22      | 0.909 [0.739, 0.981]  |

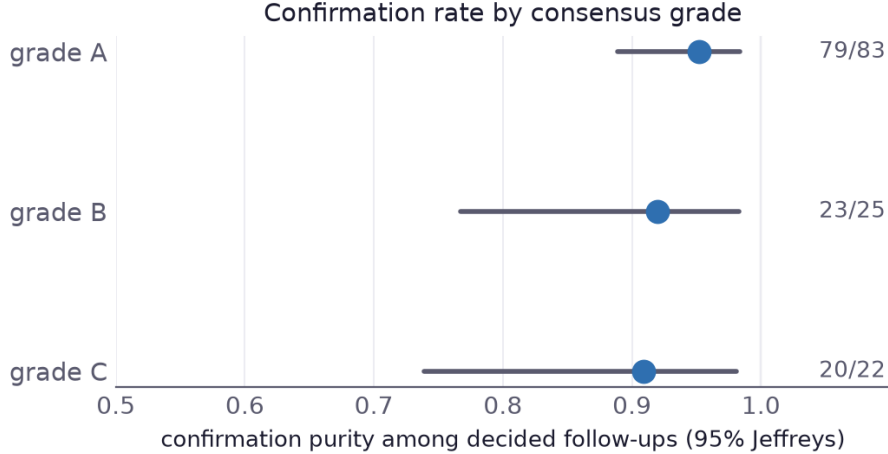
**Table 2:** The program’s first confirmation-rate-per-grade table, over all published follow-up (Foundry I–IV, AGEL DR2, HSC×DESI, SuGOHI spectroscopy). “Decided” = confirmed + refuted. Follow-up targets were curated — mostly best-first — so purities are upper bounds and the grade comparison inherits each campaign’s selection function (§7).

of a HOLISMOKES blind set had zero score dispersion among five graders ([Cañameras et al., 2021](#)); only 25.8% of another team’s grade-A DR9 candidates received A or B from this program’s graders ([Storfer et al., 2024](#)); the Euclid Q1 panel is decisive on only  $\sim 38\%$  of its candidates ([Euclid Collaboration et al., 2025](#)); and all eleven confirmed SLACS lenses embedded in DES-quality imaging were collectively judged non-lenses by 55 experts ([Rojas et al., 2023](#)).

## 5 Grade-stratified confirmation rates

Reliability is not validity: graders could disagree yet still, in aggregate, select real lenses. We therefore assembled every published per-target follow-up outcome for the program’s candidates into normalized truth tables — 551 targets across seven campaigns — and cross-matched them (radius  $2''$ ) against the unified graded-candidate table (4,354 unique candidates from [Huang et al. 2020, 2021](#); [Inchausti Reyes et al. 2025](#); [Storfer et al. 2024](#)). Each matched candidate gets one outcome by precedence (confirmed > refuted > lens-z-only > inconclusive > pending); no candidate is both confirmed and refuted by different campaigns. Table 2 and Figure 2 give the result: 162 unique graded candidates have follow-up, 130 are decided, and confirmation purity among the decided is 95.2% (A), 92.0% (B), 90.9% (C) — high everywhere, flat in grade within the errors.

Two readings matter. First, **grade A is not infallible**: four grade-A candidates are spectroscopically refuted by VLT/MUSE ([Lin et al., 2025](#)) — a spiral arm at the galaxy’s own redshift, a tidal tail between merging galaxies at one redshift, a blue arc at exactly the deflector’s redshift, and an “arc” that is a *foreground* object. Every one is the morphology-only failure mode: an arc-shaped feature whose redshift, not its shape, is disqualifying. Grades assigned from  $1.3''$ -seeing pixels cannot see redshift. Second, **grade C is not mostly junk** — 20 of 22 decided C’s are real lenses. This



**Figure 2:** Confirmation purity by consensus grade among decided follow-ups, with 95% Jeffreys intervals. The three grades are statistically indistinguishable on this follow-up mix.

complements the Part I Euclid result (53% of DESI grade-C candidates re-grade to A/B at  $0.1''$ ): C tracks *resolution-limited visibility*, not lens likelihood, on the pool that teams chose to follow up.

For completeness: the six AGEL false positives (Barone et al., 2025) all trace to DES/Jacobs-selected candidates — none lies within arcminutes of any candidate in this program’s catalogs — and Foundry II’s grade-agnostic DESI Secondary Target Program (2,157 systems) remains the future, selection-clean version of Table 2 (Huang et al., 2025).

## 6 The machine-parity targets these results set

These measurements convert “performs as well as human experts” into two falsifiable, pre-registrable endpoints for any automated grader of DESI-resolution cutouts:

- E1 — agreement parity.** Score the machine’s grades against the published consensus with  $\kappa_{\text{qw}}$ . The human anchors are now measured: 0.776 (an individual team grader, upper bound), 0.42 (the two graders’ mutual agreement), 0.29 (an independent expert team). A defensible claim of agreement-parity is  $\kappa_{\text{qw}} \geq 0.29$  with a CI excluding 0 (matches an independent expert team); a strong claim is  $\kappa_{\text{qw}}$  compatible with 0.776 (indistinguishable from a team member). LENSJUDGE Part I’s DESI-resolution graders sat at  $\kappa \approx 0$  — the gap to close is now quantified.
- E2 — truth parity.** On candidates with decided follow-up, compare machine score and human grade (as an ordinal score) as paired predictors of confirmed-vs-refuted, via non-inferiority on AUC with a pre-registered margin. The human operating points from Table 2 — e.g. A/B selection at  $\sim 94\%$  decided purity on the curated mix — are the benchmarks. Today’s decided sample ( $n = 130$ , only 8 refutations) supports estimation but is thin for a verdict at small margins; the forthcoming grade-agnostic Foundry program and the Euclid Q1/DR1 truth sets (hundreds of space-resolution adjudications inside the Legacy footprint) are what will power it properly.

Both endpoints need the machine to be given the graders’ actual task: Paper II–DR10 graders saw multi-band composites with viewer context and redshift metadata, not bare 101-px pixels — an input asymmetry any fair comparison must remove.

## 7 Caveats

(i) Truncation: the Paper II pairs exist only for accepted candidates; all reliability numbers are upper bounds. (ii) Rater anonymity: unordered pairs support interchangeable-rater statistics only; per-grader bias is unrecoverable. (iii) Selection: follow-up campaigns targeted their best candidates (Foundry I explicitly so, and its 51/51 confirmation rate should be read as a property of that curation); purities are upper bounds and cross-grade comparisons inherit the selection. (iv) Truth precedence: “confirmed” requires both lens and source redshifts (or HST morphology per Foundry I); source-only AGEL systems are held out as inconclusive. (v) The cross-team  $\kappa_{\text{qw}}$  compares different rubrics, renderings, and (for Euclid) resolutions; part of that gap is real information difference, so the same-regime SuGOHI comparison is the cleaner anchor. (vi) One grading pass per pair: intra-rater noise (Rojas’s 73%) is not separable from inter-rater disagreement here.

## 8 Reproducibility

All inputs are published catalogs. `parity/fetch_vizier_paper2.py` retrieves and validates the Paper II table (row parity check; 100% join against the local reproduction catalog); `parity/intergrader_stats.py` computes Table 1; `eval/crossmatch_external.py` rebuilds the master candidate table and the SuGOHI/Euclid crossmatches (104 and 24 matches, reproducing Part I); `eval/human_ceiling.py` computes the cross-team  $\kappa_{\text{qw}}$ ; `parity/data/external/*.csv` hold the seven normalized per-target truth tables (each machine-parsed from the papers’ own LaTeX/VizieR/Zenodo sources, with per-row provenance); `parity/grade_purity_table.py` produces Table 2; and `parity/human_baseline_summary.py` assembles the combined baseline table. Figures: `parity/make_baseline_figures.py`.

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